Replicating Bartels with the CES: Party Identification, 2008-2024

Josh Arden

2026-10-18

1. Introduction

This write-up replicates Figures 1 and 2 from Bartels (2000), "Partisanship and Voting Behavior, 1952-1996," using the Cumulative CES (Kuriwaki 2025) for 2006-2024 instead of the ANES. Because the CES starts in 2006, it cannot reproduce Bartels' 1952-1996 years. It can only show whether his patterns hold in recent presidential elections (2008, 2012, 2016, 2020, 2024).

Figure 1 shows the distribution of party identification (strong, weak, leaning independent, pure independent) among all respondents. Figure 2 shows the share of strong or weak identifiers among voters and nonvoters.

2. Method and Code

The project is split into stages that a master script runs in order: read, clean, then draw each figure. Each stage after reading is a function that takes data in and hands data back, so the figure functions only use the data they are given.

2.1 Reading

```
# read in the Cumulative CES, 2006-2024
raw_data <- read_dta(file.path(raw_dir, "cumulative_2006-2024.dta"))
```

2.2 Cleaning

`clean_cces()` keeps the columns the figures use, keeps presidential years, and recodes party ID and turnout. It does not drop any respondents, so each figure script defines its own sample.

Turnout is self-reported (`voted_turnout_self`) because validated 2024 turnout is not yet released in the cumulative file (it is planned for Version 11). Leaners are coded as non-partisan, following Bartels' footnote 2.

```
clean_cces <- function(raw_data) {
  raw_data |>
    select(year, case_id, pid7, voted_turnout_self,
           weight, weight_post, voted_pres_party) |>
    filter(year %in% c(2008, 2012, 2016, 2020, 2024)) |>
    mutate(
      pid_cat = case_when(
        pid7 %in% c(1, 7) ~ "strong",
        pid7 %in% c(2, 6) ~ "weak",
        pid7 %in% c(3, 5) ~ "lean_ind",
        pid7 == 4         ~ "pure_ind"
      ),
      partisan = case_when(
        pid7 %in% c(1, 2, 6, 7) ~ 1,
        pid7 %in% c(3, 4, 5)    ~ 0
      ),
      voter = case_when(
        as_factor(voted_turnout_self) == "Yes" ~ 1,
        as_factor(voted_turnout_self) == "No"  ~ 0
      )
    )
}

clean_data <- clean_cces(raw_data)
```

Respondents with "Not Sure," "Don't Know," or missing party ID have no `pid_cat` and are dropped from both figures. Respondents with missing turnout are dropped from Figure 2 only.

```
# how many respondents each figure keeps, by year
clean_data |>
  group_by(year) |>
  summarise(all_rows        = n(),
```

```
            fig1_sample     = sum(!is.na(pid_cat)),
            fig2_sample     = sum(!is.na(pid_cat) & !is.na(voter)))
```

```
# A tibble: 5 x 4
   year all_rows fig1_sample fig2_sample
  <dbl>    <int>       <int>       <int>
1  2008    32800       31853       26327
2  2012    54535       52052       41125
3  2016    64600       62479       48120
4  2020    61000       58631       47725
5  2024    60000       58738       46545
```

2.3 Figure 1 function

```
make_fig1 <- function(clean_data) {

  fig1_data <- clean_data |>
    filter(!is.na(pid_cat)) |>
    group_by(year, pid_cat) |>
    summarise(n = sum(weight), .groups = "drop") |>
    group_by(year) |>
    mutate(
      share = n / sum(n),
      panel = case_when(
        pid_cat %in% c("strong", "weak")       ~ "A",
        pid_cat %in% c("lean_ind", "pure_ind") ~ "B"
      ),
      pid_cat = factor(pid_cat,
                       levels = c("strong", "weak", "lean_ind", "pure_ind"),
                       labels = c("Strong Identifiers", "Weak Identifiers",
                                  "Independent Leaners", "Pure Independents"))
    ) |>
    ungroup()

  election_years <- c(2008, 2012, 2016, 2020, 2024)

  bartels_theme <- theme_minimal(base_size = 12) +
    theme(
      panel.border      = element_rect(color = "black", fill = NA, linewidth = 0.5),
      panel.grid.minor  = element_line(linewidth = 0.2),
```

```
    panel.grid.major  = element_line(linewidth = 0.3),
    legend.background = element_rect(fill = "white", color = "grey70", linewidth = 0.3),
    plot.title        = element_text(face = "bold")
  )

base_panel <- function(d) {
  ggplot(d, aes(x = year, y = share, linetype = pid_cat, shape = pid_cat)) +
    scale_y_continuous(limits = c(0, 0.5), breaks = seq(0, 0.5, 0.1),
                       minor_breaks = seq(0, 0.5, 0.05), name = NULL) +
    scale_x_continuous(breaks = election_years, minor_breaks = election_years,
                       name = "Year") +
    geom_hline(yintercept = seq(0.1, 0.5, 0.1),
               linetype = "dotted", linewidth = 0.2, color = "black") +
    geom_vline(xintercept = election_years,
               linetype = "dotted", linewidth = 0.2, color = "black") +
    geom_line(linewidth = 0.7, color = "black") +
    geom_point(size = 2.2, stroke = 0.9, color = "black", fill = NA) +
    guides(linetype = guide_legend(ncol = 1, title = NULL),
           shape    = guide_legend(ncol = 1, title = NULL)) +
    bartels_theme
}

panel_a <- base_panel(filter(fig1_data, panel == "A")) +
  scale_linetype_manual(values = c("Strong Identifiers" = "solid",
                                   "Weak Identifiers"   = "dashed")) +
  scale_shape_manual(values = c("Strong Identifiers" = 16,
                                "Weak Identifiers"   = 1)) +
  labs(title    = "Distribution of Party Identification, 2008-2024",
       subtitle = "Proportions of CES Sample",
       caption  = "Presidential Election Years Only") +
  theme(legend.position = c(0.6, 0.2))

panel_b <- base_panel(filter(fig1_data, panel == "B")) +
  scale_linetype_manual(values = c("Independent Leaners" = "dashed",
                                   "Pure Independents"   = "solid")) +
  scale_shape_manual(values = c("Independent Leaners" = 1,
                                "Pure Independents"   = 16)) +
  theme(legend.position = c(0.4, 0.8))

(panel_a + theme(axis.title.x = element_blank(), axis.text.x = element_blank(),
                 axis.ticks.x = element_blank(), plot.caption = element_blank())) /
  (panel_b + theme(axis.title.x = element_blank()))
```

```
}
```

2.4 Figure 2 function

```
make_fig2 <- function(clean_data) {

  fig2_data <- clean_data |>
    filter(!is.na(pid_cat), !is.na(voter)) |>
    group_by(year, voter) |>
    summarise(share = weighted.mean(partisan, weight), .groups = "drop") |>
    mutate(group = if_else(voter == 1, "Voters", "Nonvoters"))

  election_years <- c(2008, 2012, 2016, 2020, 2024)

  ggplot(fig2_data, aes(x = year, y = share, linetype = group, shape = group)) +
    geom_hline(yintercept = seq(0.1, 1.0, 0.1),
               linetype = "dotted", linewidth = 0.2, color = "black") +
    geom_vline(xintercept = election_years,
               linetype = "dotted", linewidth = 0.2, color = "black") +
    geom_line(linewidth = 0.7, color = "black") +
    geom_point(size = 2.2, stroke = 0.9, color = "black", fill = NA) +
    scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, 0.1),
                       minor_breaks = seq(0, 1, 0.05), name = NULL) +
    scale_x_continuous(breaks = election_years, minor_breaks = election_years,
                       name = "Year") +
    scale_linetype_manual(values = c("Nonvoters" = "dashed", "Voters" = "solid")) +
    scale_shape_manual(values = c("Nonvoters" = 1, "Voters" = 16)) +
    guides(linetype = guide_legend(ncol = 1, title = NULL),
           shape    = guide_legend(ncol = 1, title = NULL)) +
    labs(title    = "Party Identification Among Presidential Voters and Nonvoters",
         subtitle = "Proportions of Strong or Weak Identifiers in CES Sample",
         caption  = "Presidential Election Years Only, Weighted") +
    theme_minimal(base_size = 12) +
    theme(
      panel.border      = element_rect(color = "black", fill = NA, linewidth = 0.5),
      panel.grid.minor  = element_line(linewidth = 0.2),
      panel.grid.major  = element_line(linewidth = 0.3),
      legend.position   = c(0.2, 0.2),
      legend.background = element_rect(fill = "white", color = "grey70", linewidth = 0.3),
      plot.title        = element_text(face = "bold")
```

```
    )
}
```

3. Results

3.1 Figure 1: distribution of party identification

```
fig1 <- make_fig1(clean_data)
fig1
```

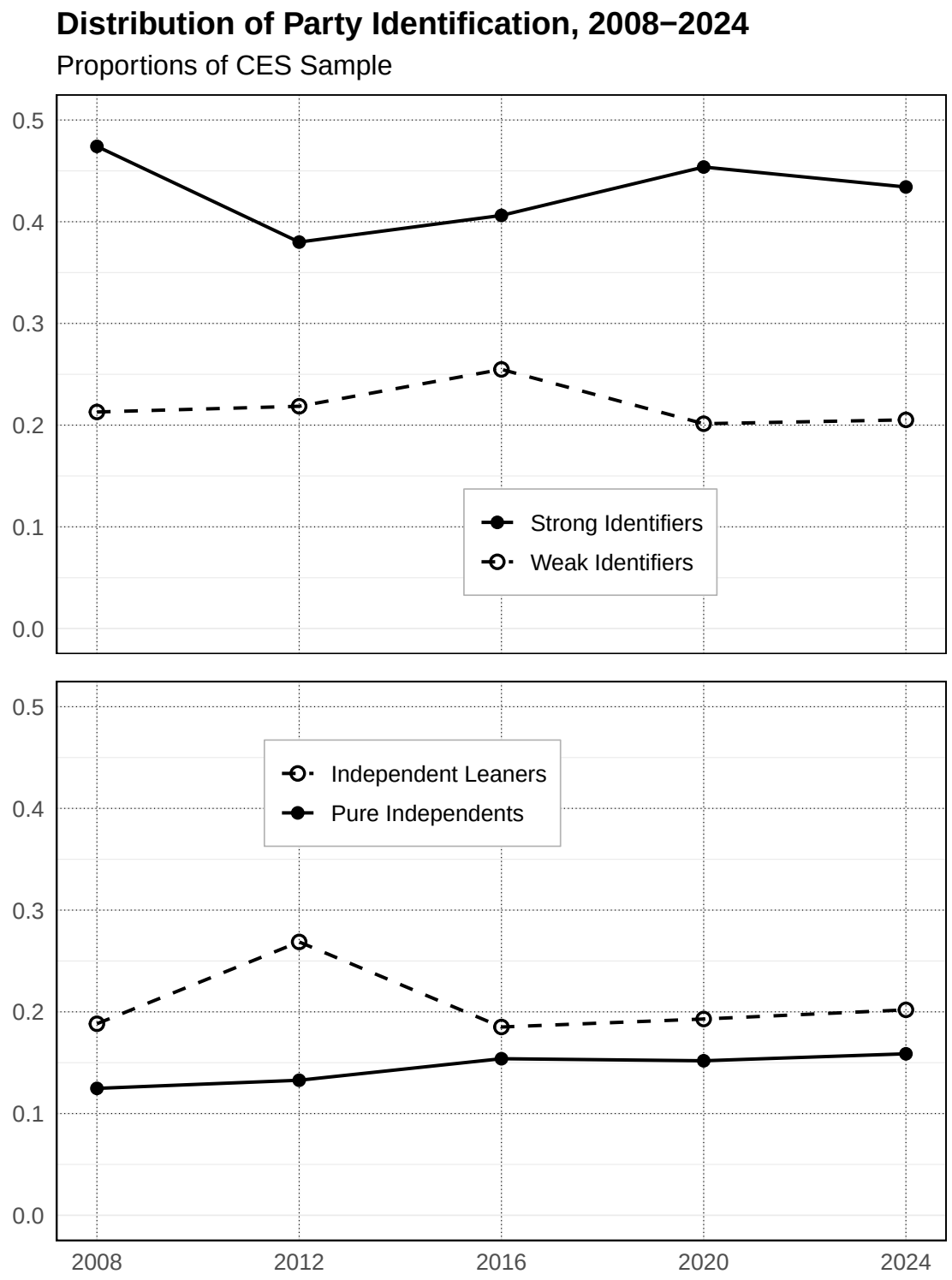


Figure 1: Distribution of party identification in the CES, presidential years 2008-2024. Weighted shares among respondents with a valid party ID.

Strong identifiers outnumber weak identifiers in every year, and leaners outnumber pure independents in every year. Between 2008 and 2012, strong identifiers fall from about 0.47 to 0.38 while leaners rise from about 0.19 to 0.27, with weak identifiers roughly unchanged. Because this is a repeated cross-section, this shows a change in the sample's composition, not individuals switching categories.

3.2 Figure 2: voters and nonvoters

```
fig2 <- make_fig2(clean_data)
fig2
```

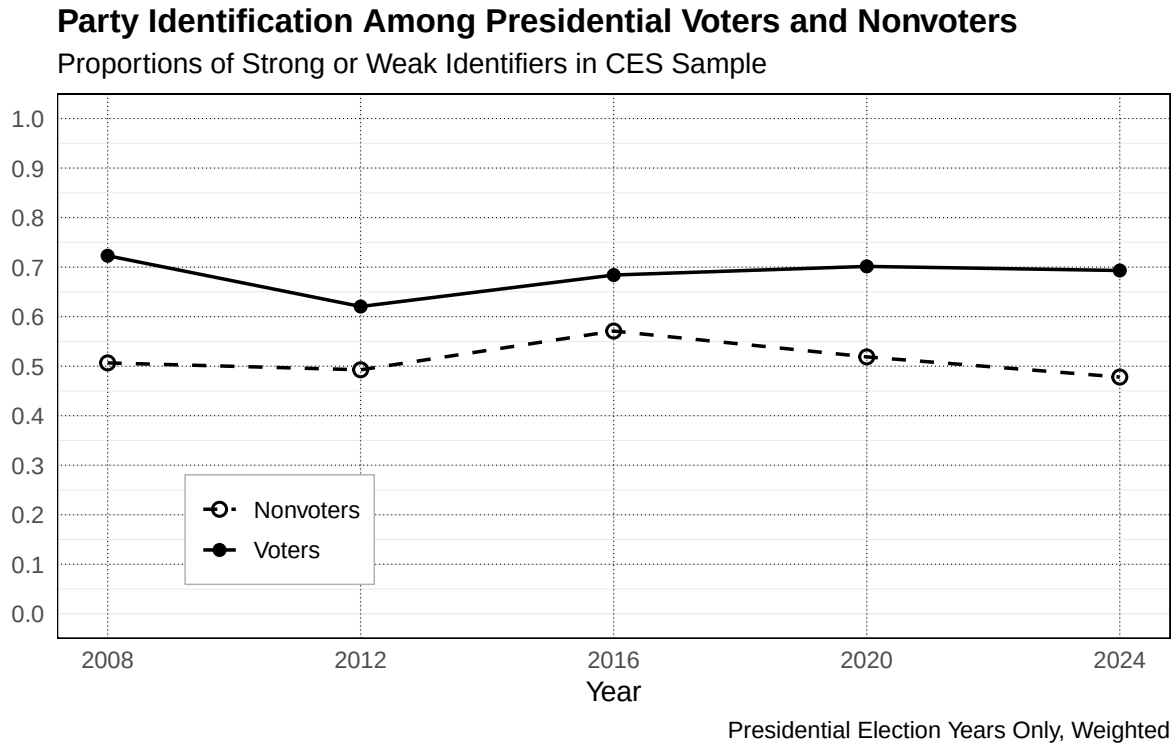


Figure 2: Share of strong or weak identifiers among presidential voters and nonvoters in the CES, 2008-2024. Turnout is self-reported.

Voters are more partisan than nonvoters in every year, by roughly 11 to 21 percentage points. The gap narrows through 2016 and then widens again, so there is no clear trend from 2008 to 2024.

Bartels reports that about 75 to 77 percent of voters were party identifiers in 1952, 1956, and 1996, and that the voter/nonvoter gap grew from about 10 points in the 1950s to about 20 points by the 1990s. The CES voter line is lower than his (about 0.62 to 0.72), but the gap in 2008 and 2024 (about 21 points) is similar to his 1990s level. I did not replicate Figure 2 with the ANES, so this comparison uses the numbers Bartels reports in his text.

3.3 Sensitivity to weighting in 2012

```
# unweighted vs. weighted shares of each party ID category, by year
clean_data |>
  filter(!is.na(pid_cat)) |>
  group_by(year, pid_cat) |>
  summarise(unweighted = n(), weighted = sum(weight), .groups = "drop") |>
  group_by(year) |>
  mutate(unweighted = unweighted / sum(unweighted),
         weighted   = weighted / sum(weighted)) |>
  print(n = 20)
```

```
# A tibble: 20 x 4
# Groups:   year [5]
    year pid_cat  unweighted weighted
   <dbl> <chr>         <dbl>    <dbl>
 1  2008 lean_ind      0.200    0.188
 2  2008 pure_ind      0.102    0.125
 3  2008 strong        0.497    0.474
 4  2008 weak          0.201    0.213
 5  2012 lean_ind      0.225    0.269
 6  2012 pure_ind      0.119    0.133
 7  2012 strong        0.449    0.380
 8  2012 weak          0.207    0.219
 9  2016 lean_ind      0.189    0.185
10  2016 pure_ind      0.168    0.154
11  2016 strong        0.396    0.406
12  2016 weak          0.247    0.255
13  2020 lean_ind      0.201    0.193
14  2020 pure_ind      0.151    0.152
15  2020 strong        0.447    0.454
16  2020 weak          0.200    0.201
17  2024 lean_ind      0.203    0.202
18  2024 pure_ind      0.134    0.159
```

```
19  2024 strong         0.467    0.434
20  2024 weak           0.196    0.205
```

The 2012 dip in strong identifiers and the spike in leaners are much larger in the weighted estimates than in the raw responses. Unweighted, strong identifiers fall from 0.50 to 0.45 and leaners rise from 0.20 to 0.23. Weighted, strong falls from 0.47 to 0.38 and leaners rise from 0.19 to 0.27. Weighting changes the estimates little in 2016 and 2020, so the 2012 result is sensitive to weighting. I could not determine why.

4. Discussion

4.1 Challenges

The hardest part was understanding how the master script passes information between stages. I had to learn that `source()` only defines a function unless the script runs code, that the function has to be called to produce an object, and that the parameter name inside a function is a stand-in for the object the master hands it. Building file paths with `here()` and `file.path()` and saving figures into the project's `output` folder took extra effort, but it made the project reproducible on another computer.

I also had to decide how to handle 2024 turnout. Validated turnout for 2024 is not yet released in the cumulative file, so I used self-reported turnout, which is available for every year but likely overstates voting. Other limits are that the CES covers only 2006-2024, so Bartels' years cannot be reproduced, and that post-election weights are missing for some years, so I used the pre-election weight even though turnout comes from the post-election wave. The 2012 weighting result in Section 3.3 was the most analytical problem I ran into.

4.2 Differences from the ANES replication

I compared the CES to my ANES replication of Figure 1 (I did not replicate Figure 2 with the ANES). My ANES replication matches Bartels' check: strong identifiers are about 24 percent in 1976 and about 30 percent in 1996, close to his 24 and 31 percent.

The two surveys agree closely in 2020 and 2024: strong identifiers outnumber weak, at roughly 0.43 to 0.45 in the CES and 0.41 to 0.44 in the ANES. They diverge in 2008 and 2012. The CES shows many more strong identifiers in 2008, and its weighted 2012 dip in strong identifiers does not appear in the ANES, which supports treating that dip as sensitive to weighting. In 2024 the ANES shows far fewer pure independents (about 7 percent) than the CES (about 16 percent).

The surveys also differ in design (an online matched sample versus the ANES) and in how missing responses are handled: I kept them in the ANES denominator but dropped them in the CES. For that reason I compare patterns, not exact levels.